

Qatar-4b

R-2209

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_241129 · created 2026-08-14T01:27:48+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460643.7025 +/- 0.0058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.036
Transit depth (Rp/Rs)^2	3e-06 +/- 0.0012 [%]
Orbital Inclination (inc)	89.11 +/- 2.19
Airmass coefficient 1 (a1)	1.206 +/- 0.034
Airmass coefficient 2 (a2)	-0.179 +/- 0.099
Scatter in the residuals of the lightcurve fit is	2.84 %
Best Comparison Star	#1 - [361, 415]
Optimal Method	PSF photometry
Transit Duration (day)	0.0748 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.0 ± 0.036 vs 0.138 (deviation 3.83σ)
Duration fit vs published	0.0748 d vs 0.0892 d (deviation 2.18σ)
Mid-time O-C	-5.4 min
Depth SNR	0.0
Residual scatter	2.84 %

Light curve

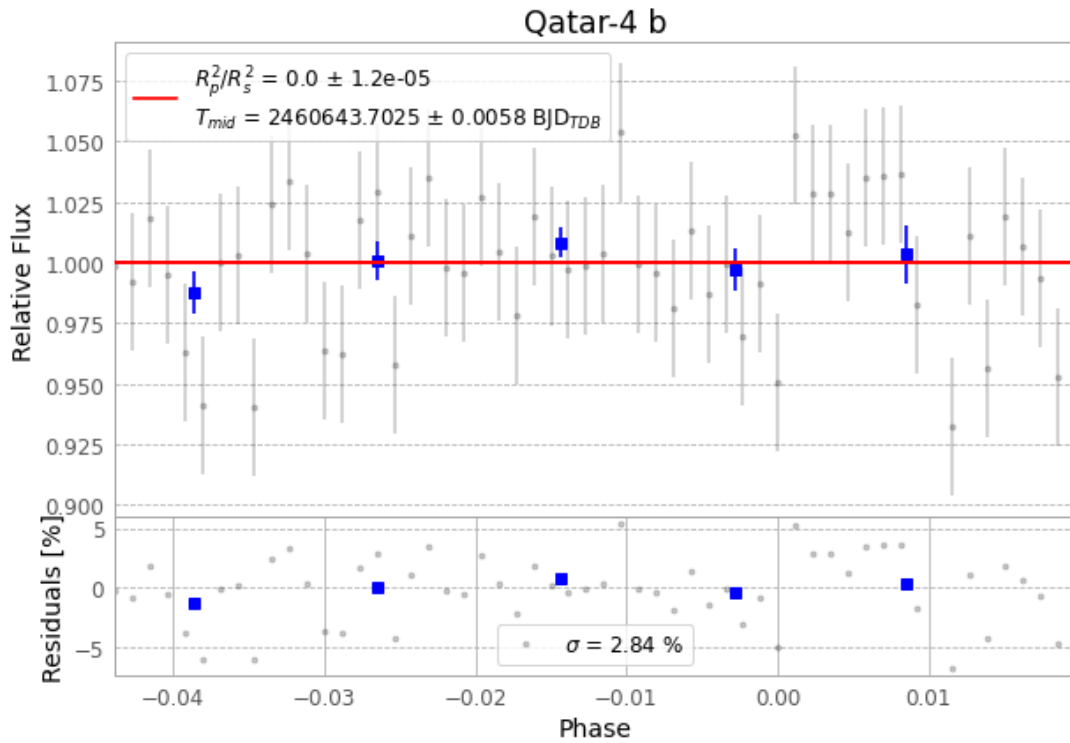


Figure 1 — FinalLightCurve_Qatar-4 b_2024-11-28.png (SHA-256 c6dd96064767db0d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [395, 247], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 85bde47dd910a35f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

56 FITS frames (37 MB), Qatar-4241129025115.FITS → Qatar-4241129053615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-241129.log (1722b9fac384...), FinalLightCurve_Qatar-4 b_2024-11-28.png (c6dd96064767...), FinalLightCurve_Qatar-4 b_2024-11-28.csv (86e5ffc4ae6e...)

Compute resources

Wall time 91.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 01:27Z.